

Land Acquisition System — Stakeholders, Bottlenecks, Modules & Data Model

Part 2 of the domain research series — read after the Process Stages document

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Purpose and Scope

This document continues the domain research started in the process-stages document. Where that document explained *what happens* at each of the nine stages of land acquisition, this one covers *who is involved, where things actually go wrong in practice, how the problem statement's required modules map onto the real process, and what the underlying data model should look like*.

Part A: Stakeholders — Who Actually Does What

Requiring Body The department or agency that needs the land — a highway authority, Railways, a state industrial corridor authority, an irrigation department. Initiates the process by identifying land and requesting acquisition. Critically, the Requiring Body must **deposit the R&R cost with the Collector before the Declaration (Stage 5) can be issued** — a funding gate that is a natural, concrete trigger condition for a workflow system: no deposit recorded, no declaration allowed to proceed.

Collector (the statutory term — defined as “the Collector of a revenue district,” and includes Deputy Commissioners or any officer specially designated to perform Collector functions under the Act) The central executing authority at district level. Conducts the land survey and measurement (Stage 5), determines the compensation award (Stage 6), reviews the draft R&R scheme before forwarding it upward, and holds the power to take possession (Section 38). This is effectively the primary workflow-owner role across most of the process.

Administrator for Rehabilitation & Resettlement (Section 43) A role distinct from the Collector, dedicated to the R&R side of the process: prepares the draft R&R scheme, organizes public hearings on it, and submits it to the Collector for review. This role is what makes R&R a genuinely separate workflow track rather than a sub-task buried inside the Collector's job.

Commissioner for Rehabilitation & Resettlement (Section 44) Sits above the Administrator. Approves the R&R scheme once the Collector forwards it with recommendations, then publishes the approved scheme — in the local language, uploaded to the government website, and displayed at the Gram Panchayat, Municipality, Collector's office, Sub-Collector's office, and Tehsil office. In practice this role is often held by a Revenue Divisional Commissioner (RDC). This is a state-level approval gate sitting above the district level.

R&R Committee (project level) Constituted **only when the acquired land is 100 acres or more**, chaired by the District Collector. Monitors and reviews implementation of the R&R scheme after approval, and conducts post-implementation social audits in consultation with the Gram Sabha (rural areas) or Municipality (urban areas). This is a conditional workflow branch — it should only appear in a system's process flow above a defined size threshold.

Independent Expert Group (Stage 3) Reviews the Social Impact Assessment and can recommend cancelling the project outright. Distinct from the R&R Committee — operates earlier in the timeline and is focused on the acquisition decision itself, not on R&R execution.

State Government Commissions the Social Impact Assessment and frames state-specific rules under the Act — each state maintains its own RFCTLARR Rules (for example, Odisha RFCTLARR Rules 2016). This means genuine, legitimate variation in implementation exists between states, which is precisely why a lack of a unified national system creates real problems.

Central Ministries Set overall policy. In this context, the Ministry of Rural Development / Department of Land Resources is the body that would own a national platform of this kind.

Gram Sabha / Panchayat / Municipality Consulted during the Social Impact Assessment (Stage 2) and during R&R scheme public hearings; also consulted by the R&R Committee for post-implementation social audits. A recurring touchpoint across multiple stages, not a single formality.

Land Acquisition and Rehabilitation Tribunal Hears disputes over compensation or rehabilitation (the dispute path branching off Stage 6). Appeals from the Tribunal go to the District Collector, and further appeals go to the High Court.

Affected Persons (landowners, co-owners, agricultural laborers, and other livelihood-dependent families) The people the process is ultimately about, but structurally the least “powerful” stakeholder in terms of system access — mostly recipients of notifications and compensation rather than initiators of workflow actions.

Structural takeaway: there are genuinely **two parallel approval chains** running through this process, not one — the *acquisition chain* (Requiring Body → Collector → Award) and the *R&R chain* (Administrator → Collector review → Commissioner approval → R&R Committee, conditionally). Any role-based access model needs at minimum: Requiring Body user, District Collector, R&R Administrator, R&R Commissioner (state level), Central Ministry viewer, and a read-only citizen-facing view for affected persons to check their own status.

Part B: Where the Real Bottlenecks Are

This section is grounded in parliamentary and audit findings, not assumption — the problem statement’s claim of “fragmented systems, delays, limited transparency” has specific, documented, quantified real-world evidence behind it.

Land acquisition is the single largest cause of infrastructure project delays in India. A 2025 parliamentary standing committee report on the Ministry of Road Transport and Highways found that **35% of stalled highway projects** were delayed due to land acquisition disputes — driven specifically by inaccuracies in land records, resistance from local stakeholders, and prolonged compensation negotiations. Separately, government data reviewed under the PRAGATI (Pro-Active Governance and Timely Implementation) mechanism confirms land acquisition as the largest single blocker among all categories — ahead of environmental clearances, forest clearances, and utility shifting.

Federal complexity compounds the fragmentation. Land is a State subject under India’s constitutional structure, meaning each state frames its own RFCTLARR Rules. This produces legitimately uneven implementation nationwide — not a flaw to “fix” so much as a constraint any national system must accommodate rather than assume away.

A concrete, quantified example — the Bangalore Metro (BMRCL) audit: A CAG performance audit (Report No. 7 of 2026) found that incorrect land estimations and acquisition delays increased land acquisition costs by **₹6,603 crore** across two project phases. Separately, delays in issuing the Final Notification beyond a prescribed 270-day window resulted in an **additional ₹186.86 crore paid purely as 12% statutory interest** — money spent solely because of delay, not because of the land’s value. This is a real, audited instance of exactly the “interest accrual” mechanic already identified in Stage 6 and Stage 7 of the process document — proof that tracking a project against its statutory deadline is not a theoretical feature but something with a demonstrated financial consequence attached.

Named structural causes across multiple sources, consistently repeated: - Lengthy Social Impact Assessment studies and public hearings - Consent-threshold difficulties for private/PPP projects - Litigation over compensation valuation, consent validity, and rehabilitation adequacy - Inaccurate or outdated land records feeding into disputes over ownership and valuation - Social resistance rooted in distrust, fear of livelihood loss, and displacement concerns - Delayed compensation disbursement, which itself compounds into further delay via the statutory interest mechanism (this is the one bottleneck that is self-reinforcing — a delay literally causes a bigger delay-cost, not just a longer wait)

What this means for system design: the strongest, most defensible framing for a “predictive” or “risk” feature is not a generic ML buzzword — it is tracking projects against their **actual statutory deadlines** (the 270-day-style notification windows, the 12-month award deadline, the 5-year lapse deadline) and flagging projects approaching them, since these deadlines have real, audited financial consequences attached when missed. This is closer to a compliance/deadline-tracking engine than a predictive model, and it is more credible precisely because it is grounded in provable statutory triggers rather than an opaque prediction.

Part C: Mapping the Problem Statement’s Scope of Study to System Modules

The problem statement’s own “Scope of Study” table effectively specifies the module list for the system. Below, each

listed aspect is connected back to the specific process stage(s) and stakeholder(s) it corresponds to, so the module list is grounded in the real process rather than treated as an abstract checklist.

Scope of Study Aspect (from PS)	Corresponding Process Stage(s)	Corresponding Stakeholder(s)
Existing System study	All stages — baseline understanding	All
Workflow Digitization (proposal → scrutiny → approvals → notifications → awards → compensation → possession → closure)	Stages 1, 3, 5, 6, 7, 8	Requiring Body, Collector
GIS & Geotagging	Stage 5 (survey/measurement) primarily; Stage 1 (parcel identification)	Collector
Data Standardization	Cuts across all stages — this is precisely the fragmentation problem named in Part B	State Government (source of variation)
Dashboard & Analytics	Cuts across all stages, aggregating status	Central Ministries, State Government (viewers)
Integration (with land records, cadastral maps, financial systems)	Stage 5 (ownership claims), Stage 6 (compensation calculation inputs)	Collector
Document Management	Stages 1, 2, 5 (notifications, SIA reports, ownership documents)	All
Monitoring & Alerts	Stage 6 (12-month award deadline), Stage 8 (5-year lapse deadline)	Collector, Central Ministries
Reporting	Cuts across all stages	Central Ministries, State Government
Security & Governance (role-based access, audit trails)	Cuts across all stages	All — access differs by role
Scalability (multilingual, nationwide)	Reflects Part B’s federal-complexity finding directly	State Government

This mapping shows the module list is not arbitrary — each item traces back to a genuine point of friction or a genuine stakeholder need already identified in Parts A and B.

Part D: Data Model — Core Entities

This is a starting entity list derived directly from the process, stakeholders, and problem statement’s explicit dashboard requirements — not a generic CRUD schema.

Project project_id, title, requiring_body, acquisition_route (RFCTLARR / Railway Act / Other), project_type (Government / Private / PPP — determines whether Stage 4 consent applies), current_stage, state, district, created_at

LandParcel parcel_id, project_id (FK), survey_number, khasra_number, khata_number, village, tehsil, district, area, land_classification (including a multi-crop-irrigated flag per the Stage 2 safeguard), geo_boundary (spatial data), total_area_acres (rolled up per project, relevant to the 100-acre R&R Committee threshold)

AffectedPerson person_id, parcel_id (FK, may link to multiple parcels), role (Owner / Co-owner / Agricultural Laborer / Other Livelihood-Dependent), sc_st_status (for the Stage 9 additional-protection flag), ownership_claim_details, compensation_claim_amount

Notification notification_id, project_id (FK), type (Preliminary Section 11 / Declaration Section 19), publication_date, gazette_reference, newspaper_reference, notice_board_proof, objection_window_deadline

SIA_Record sia_id, project_id (FK), report_document, public_hearing_log, expert_group_outcome (Approved / Modified / Rejected), expert_group_justification

ConsentRecord (only relevant where project_type is Private or PPP) consent_id, project_id (FK), affected_families_total, consents_collected, threshold_required (80% or 70%), current_percentage

Award award_id, project_id (FK), declaration_date, award_deadline (declaration_date + 12 months), award_date, market_value, asset_value, solatium_amount, area_type (rural/urban — determines multiplier), multiplier_applied, interest_accrued, final_compensation_amount, dispute_status

Disbursement disbursement_id, award_id (FK), claimant_id (FK to AffectedPerson), disbursed_amount, disbursement_date, days_delayed_past_award, additional_interest_accrued

Possession possession_id, project_id (FK), possession_date, lapse_risk_deadline (award_date + 5 years per Section 24(2)), lapse_status

RR_Scheme scheme_id, project_id (FK), administrator_id, draft_status, collector_review_status, commissioner_approval_status, publication_status, rr_committee_required (true if total_area_acres >= 100)

RR_Entitlement entitlement_id, person_id (FK to AffectedPerson), entitlement_type (Housing / Transport Allowance / Employment / Annuity), status, disbursement_date

User / Role user_id, role (Requiring Body / Collector / RR Administrator / RR Commissioner / Central Ministry Viewer / Affected Person — read-only), jurisdiction (state/district scope)

Why this model differs from a generic land-records schema: the separation of AffectedPerson from a simple “landowner” field, the explicit award_deadline and lapse_risk_deadline fields tied to statutory sections rather than generic “due dates,” and the conditional rr_committee_required flag are all details that only emerge from understanding the actual legal process — not from looking at the problem statement text alone.